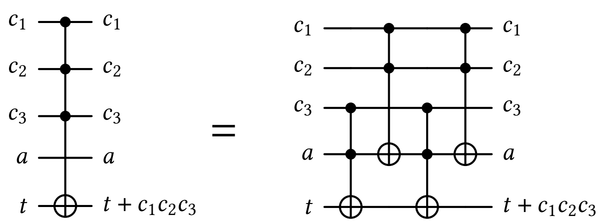

Fig. 12. Circuit implementing 3-controlled X gate.

$$
\begin{aligned}
S \triangleq\ & \{\bar{c}, t \mapsto |\bar{C}, T\rangle\} \\
& \texttt{qalloc}(a); \\
& Toffoli(c_3, a, t); \\
& Toffoli(c_1, c_2, a); \\
& Toffoli(c_3, a, t); \\
& Toffoli(c_1, c_2, a); \\
& \texttt{qfree}(a); \\
& \{\bar{c}, t \mapsto |\bar{C}, T + C_0C_1C_2\rangle\}
\end{aligned}
$$

Fig. 13. Qwhile-hp program for the 3-controlled X gate.

8.1 Programs with Dirty Ancilla Qubits: In-place Addition Circuit and MCX Gate

The first example is the one introduced in Section 2, where the circuit shown in Fig. 1 functions as an in-place constant adder computing the last bit of $r = a + (011)_2$. The quantum circuit can be implemented in Qwhile-hp as follows:

$$
\begin{aligned}
S \quad \triangleq \quad & \texttt{qalloc}(g_0); \texttt{qalloc}(g_1); CNOT[g_1, a_2]; CNOT[a_1, g_1]; X[a_1]; Toffoli[g_0, a_1, g_1]; \\
& CNOT[a_0, g_0]; Toffoli[g_0, a_1, g_1]; CNOT[g_1, a_2]; Toffoli[g_0, a_1, g_1]; \\
& CNOT[a_0, g_0]; Toffoli[g_0, a_1, g_1]; X[a_1]; CNOT[a_1, g_1]; \texttt{qfree}(g_0); \texttt{qfree}(g_1)
\end{aligned} \tag{1}
$$

The correctness specification for S is formulated as follows:

$$\forall \bar{C} \in \{0,1\}^3. \ \models \{\bar{a} \mapsto |\bar{C}\rangle\langle\bar{C}|\}\ S\ \{\bar{a} \mapsto |f(\bar{C})\rangle\langle f(\bar{C})|\}$$

where $f(C_0, C_1, C_2) = C_0, C_1, C_2 + C_1 + C_0 + C_0C_1$ and $\bar{a} = a_0, a_1, a_2$. This specification can be derived using the proof system in Fig. 9. Furthermore, the body of S, which is the subprogram of S without the `qalloc` and `qfree` statements (highlighted in red in Eq. (1)) at the beginning and the end, correctly uses dirty qubits g_0 and g_1. This is established by Corollary 7.1 and the following valid triples where $g_0', g_1' \notin \bar{a} \cup \{g_0, g_1\}$:

$$\models \{\bar{a} \mapsto I * g_0, g_0' \mapsto |\Phi\rangle\langle\Phi| * g_1, g_1' \mapsto |\Phi\rangle\langle\Phi|\}\ S.body\ \{g_0, g_0' \mapsto |\Phi\rangle\langle\Phi| * g_1, g_1' \mapsto |\Phi\rangle\langle\Phi| * \top\}$$

Therefore, we can confidently borrow dirty qubits g_0 and g_1 from any other part of computation when they are allocated at the beginning of S. Their original states will be restored before they are freed at the end of S.

Next, we verify an implementation of the *Multiple-Controlled X (MCX) Gate* with dirty ancilla qubits. Fig. 12 demonstrates the circuit that implements a 3-controlled X gate using 4 Toffoli gates and 1 dirty qubit. The correctness specification is represented by the triple:

$$\forall \bar{C} \in \{0,1\}^3, T \in \{0,1\}.\ \{\bar{c}, t \mapsto |\bar{C}, T\rangle\}\ S\ \{\bar{c}, t \mapsto |\bar{C}, T + C_0C_1C_2\rangle\}$$

where $\bar{c}, t \mapsto |\bar{C}, T\rangle$ is shorthand for $\bar{c}, t \mapsto |\bar{C}, T\rangle\langle\bar{C}, T|$, and the program S is shown in Fig. 13. Again, we can verify that the body of S uses the dirty ancilla qubit a correctly, using Corollary 7.1 and the validity of

$$\{\bar{c}, t \mapsto I * a, a' \mapsto |\Phi\rangle\langle\Phi|\}\ S.body\ \{a, a' \hookrightarrow |\Phi\rangle\langle\Phi|\} \text{ for } a' \notin \{c_0, c_1, c_2, t, a\}.$$